

Karthik Raja KANDAVEL

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AI Engineer building production RAG, agentic, and LLM systems end-to-end: hybrid retrieval, reranking, evaluation, and cloud deployment. Skilled in Python, ML fundamentals, and turning research prototypes into reliable, observable products.

Education

EURECOM, France Master of Science in Computer Science (Specialisation in Data Science)	Sept 2024 - Present
Sri Sairam Institute of Technology, India B.E in Computer Science and Engineering	Jun 2018 - Jun 2022

Technical Skills

- **AI/NLP & RAG:** LLMs, Agentic RAG, Hybrid Retrieval (BGE-M3, BM25), Reranking, Query Rewriting
- **ML Fundamentals:** PyTorch, XGBoost, Scikit-learn, Model Evaluation, Experimentation, SHAP
- **Programming:** Python (FastAPI), SQL (BigQuery, PostgreSQL), Bash, Docker, Git
- **Data & Vector Infra:** ChromaDB, ETL Pipelines, Knowledge Graphs, Data Integrity, BigQuery
- **Cloud & MLOps:** GCP (Cloud Run, Pub/Sub), Azure (AZ-900), CI/CD, Model Monitoring

Professional Experience

Data Science & ML Engineering Intern (R&D to Production) Faradae	Lyon, France Mar 2026 - Present
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- **Sole R&D-to-production bridge:** Collaborated with a PhD optimisation expert to transition a research MILP model into a live revenue-generating system, owning cloud deployment, testing, and go-live independently.
- **Built and deployed the full MLOps pipeline** (Docker containers on GCP: Cloud Run, Pub/Sub, BigQuery), running the model every minute with layered fallback logic, achieving **99.9% reliability** in production.
- **Shipped an ML forecasting microservice:** Developed and deployed an XGBoost model with quantile uncertainty estimation, cutting forecast error by **45.1%** vs baseline, containerized as a low-latency service.
- **Closed the MLOps feedback loop:** Built automated monthly retraining with champion/challenger evaluation and drift monitoring (PSI, KS tests), triggering proactive retraining and hands-off redeployment via CI/CD.

Recommendation from CEO Thomas Lawson and CFO Michel Lerendu: [LinkedIn Post](#) 📄

Associate Business Analyst (Data Focus) Infosys LTD	Chennai, India Aug 2022 - Aug 2024
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- **Engineered automated data workflows** using SQL and Spotfire, eliminating manual reporting overhead and accelerating data-driven decision-making for client stakeholders.
- **Resolved critical data integrity failures** by diagnosing sensor-level faults and implementing a root-cause fix, restoring accuracy to month-long historical datasets.
- **Led technical reviews and mentored 3 junior developers**, collaborating in an Agile/Jira environment with daily stand-ups and sprint planning.

AI & Data Science Projects

Production-Grade Legal RAG System Link - 🌐 📄 - (<i>Faradae</i>) <i>ChromaDB, BGE-M3, BM25, Vertex AI (Reranker), Groq Llama-3.1, Dedoc, GCP</i>	Feb 2026 - Present
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- **Personal prototype to company asset:** Built a self-correcting agentic RAG system for French legal contracts, demoed to leadership, and secured GCP credits and engineering resources for productionisation to 1,800+ pages.
- **Hybrid retrieval with reranking:** Combined BM25 + BGE-M3 dense embeddings (RRF fusion) with cross-encoder reranking, achieving **MRR 0.7754** and **Recall@5 0.9638** on a 120-QA golden dataset; reranking alone lifted MRR by 14.1%.
- **Grounded generation with LLM evaluation:** Prompt-engineered Gemini 2.5 Flash for citation-accurate answers, achieving **100% absent-clause refusal** and **95.6% citation accuracy**, with retry/failover ensuring 99.9% query availability.

ProVe – Forensic Restoration of Wikidata Verification Link - 🌐 📄 <i>PyTorch, Hugging Face Transformers, Knowledge Graphs, REST APIs, Distributed Systems</i>	Oct 2025 - Feb 2026
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- **Reproducibility audit:** Uncovered a 37.5% performance gap between the original API and public models, documenting a "Modernization Trap" where dense retrieval underperformed keyword heuristics.
- **Fine-tuned a T5-base model** with custom structural tokens, achieving **95.2% verbalization accuracy** for 1,200+ Wikidata claims, in a fault-tolerant pipeline with zero data loss across multi-hour runs.

Certifications & Relevant Coursework

- **Microsoft Azure Fundamentals (AZ-900)** Feb 2021
- **EURECOM Technical Mastery:** Achieved 18.5/20 in Software Development Methodologies and 17.5/20 in Algorithmic Machine Learning; training in Optimization Theory, Deep Learning, Distributed Systems (Cloud).

Languages

English: C2 (Proficient / Bilingual) • **French:** B1 (Intermediate / Progressing) • **Tamil:** Native